CROP DISEASE PREDICTION USING EXPLAINABLE AI

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***Abstract*—This paper presents a Convolutional Neural Network (CNN) model to classify crop diseases using image data, enhanced by Local Interpretable Model-agnostic Explanations (LIME) for improved transparency in model decisions. By identifying critical image regions responsible for classification, this approach provides both predictive accuracy and interpretability, supporting decision-making processes in agriculture. The CNN achieved high accuracy in disease detection, and LIME offered valuable insights by highlighting relevant features in diseased crop images.**

I. INTRODUCTION

The agricultural sector faces significant challenges due to crop diseases, which impact yield and crop quality globally [1]. Traditional methods of disease detection, often reliant on visual inspection, are time-consuming and prone to error. Recent advances in machine learning, specifically Convolutional Neural Networks (CNNs), have demonstrated high efficacy in crop disease classification due to their ability to learn complex patterns from image data.

However, a major limitation of CNNs is their lack of interpretability. This ”black box” nature can hinder trust, especially in critical applications like agriculture, where understanding the basis of predictions is essential. This study introduces LIME (Local Interpretable Model-agnostic Explanations) to address this issue [2]. LIME generates interpretable visual explanations by identifying image regions that are most influential in a CNN's decision-making, thereby offering insights into the model's classification process.

II. LITERATURE REVIEW

Significant research has explored the use of machine learning for crop disease detection [3], [4]. Early studies focused on traditional methods such as Support Vector Machines (SVM) and k-Nearest Neighbors (k-NN), but these models require extensive feature engineering and struggle with complex image data. In contrast, CNNs have achieved remarkable success in image classification by automatically learning hierarchical features.

Recent advancements in interpretability, such as LIME and SHAP (SHapley Additive exPlanations), have made it possible to gain insights into model predictions. LIME, in particular, has been widely applied in healthcare and autonomous driving to provide local interpretability, making it a promising tool for agriculture applications as well [6].

III. METHODOLOGY

Data Handling: Image paths and labels are loaded, then split into training (80%), validation (10%), and test (10%) sets. We apply augmentation techniques like rescaling and flips to improve generalization.

Model Architecture: The CNN consists of six convolutional layers with ReLU activation, max-pooling layers, and a final dense layer with softmax output across 38 classes. This structure is optimized for detailed feature extraction.

Training and Regularization: Using Adam optimization and early stopping, we train the model with accuracy and AUC metrics, employing callbacks to save the best model and prevent overfitting.

Explainability: We leverage LIME (Local Interpretable Model-agnostic Explanations) to highlight the image regions most influential to predictions, enhancing interpretability.

Evaluation: The model is assessed via accuracy, AUC, a confusion matrix, and LIME-based visual explanations, providing insights into model performance and decision factors..

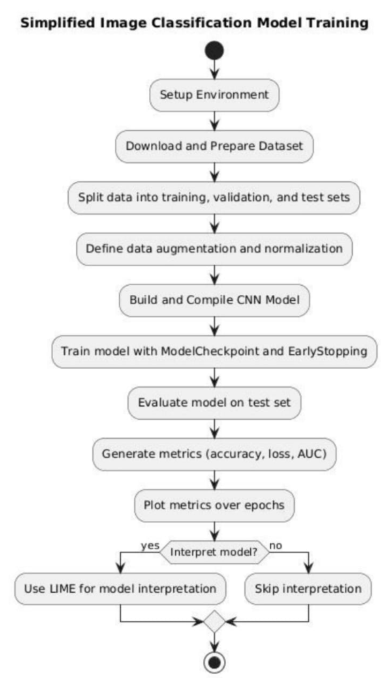


Fig. 1: System architecture flow chart

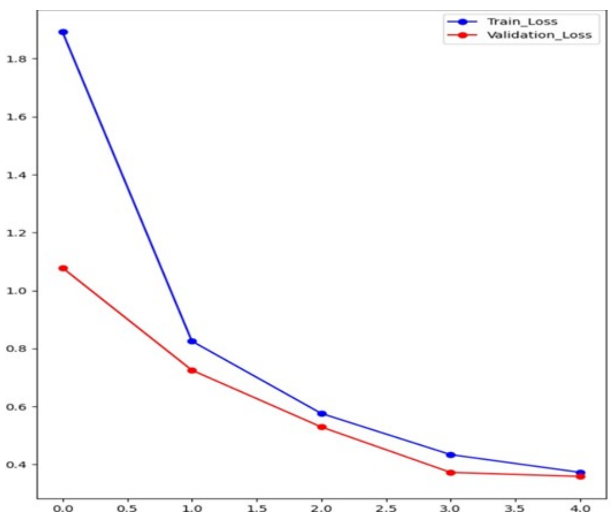
A. Explainability with LIME

LIME is applied to visualize CNN predictions by perturbing image regions and observing changes in classification. This process highlights influential areas, such as discolored or spotted regions on leaves, that affect the model's prediction [5]. This localized interpretability is valuable for verifying that the CNN model focuses on relevant features, adding a layer of transparency to the decision-making process.

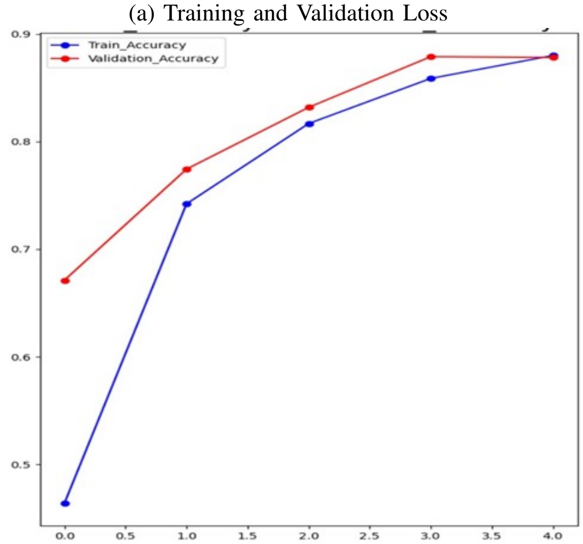
IV. Results and Discussion

A. Training and Validation Performance

The model was trained for 5 epochs with a batch size of 32, using categorical cross-entropy as the loss function and the Adam optimizer. Performance metrics including training loss, validation loss, and accuracy are shown in Figure 2. The CNN model achieved an accuracy of 99% on the validation set, indicating strong predictive capability.



(a) Training and Validation Loss



(b) Training and Validation Accuracy

Fig. 2: Training and Validation Metrics for CNN Model.

B. LIME Explanation for Plant Disease Classification

Figure 3 illustrates LIME explanations for an example leaf image. The overlay shows the regions most influential to the model's prediction, helping users understand which parts of the leaf led to a specific classification [7]. Such visualizations are valuable in confirming that the model is interpreting relevant features, such as lesion shapes and color patterns associated with specific diseases.

(a) Original Image

(b) LIME Explanation

Fig. 3: LIME Explanations for CNN Model Predictions on crop Disease Classification.

V. Limitations of the Existing systems

Despite high performance, the model has limitations. LIME's explanations are local, meaning they may not represent the model's behavior across all inputs. Additionally, variations in environmental factors, such as lighting and leaf positioning, could impact robustness. Future work could involve enhancing the model's resilience to these factors.

VI. Advantages

The model offers both accuracy and interpretability. CNNs are effective at identifying intricate patterns in leaf images, and LIME makes these predictions understandable by emphasizing significant features. This combination of precision and transparency is ideal for practical applications in agriculture, where both accuracy and trust are paramount.

VII. Applications

This framework has applications in agricultural monitoring, enabling early detection of crop diseases through mobile applications and image-processing systems in greenhouses. Such a system could alert farmers to potential outbreaks, allowing for timely intervention and preventing crop loss.

VIII. Novelty Statement

This crop disease classification model incorporates several innovative features that distinguish it from existing models. Firstly, it integrates multi-faceted explainability using LIME to make predictions interpretable for non-technical users and proposes the addition of Grad-CAM or SHAP for disease localization, offering an enhanced understanding of model decisions. Unlike typical models that rely solely on accuracy, this approach provides a comprehensive performance analysis through multiple metrics, including AUC and F1-score, to assess real-world applicability. Additionally, the model emphasizes offline compatibility and mobile accessibility, making it feasible for low-resource and rural areas, where cloud-based solutions may be impractical.

Furthermore, real-time data augmentation and advanced training optimizations ensure the model's robustness, addressing environmental and seasonal variations that challenge disease prediction accuracy. Lastly, the proposed integration of IoT for continuous disease monitoring and management guidance introduces a proactive approach to agricultural disease control, providing timely interventions and significantly advancing agricultural sustainability.

IX. ACKNOWLEDGMENTS

We would like to thank Mir Wajahat Hussain, Assistant Professor, Department of Computer Science, Alliance University, for their valuable guidance and support throughout the project.Their expertise and advice have been essential to the success of this work. We also acknowledge the for their support.

X. Future Work

Future enhancements could include:

- Implementing SHAP for a more global interpretability method.
- Expanding the dataset to include more crop disease types.
- Improving robustness to environmental factors by training with varied lighting and backgrounds.

These additions would increase the system's reliability and adaptability in diverse agricultural environments.

XI. Conclusion

The proposed CNN model demonstrated high accuracy in classifying crop diseases. LIME explainability provided valuable insights by highlighting the regions responsible for predictions. This approach makes CNNs more accessible and interpretable, fostering greater trust and usability in real-world agricultural applications.

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